EpiSegMix: a flexible distribution hidden Markov model with duration modeling for chromatin state discovery

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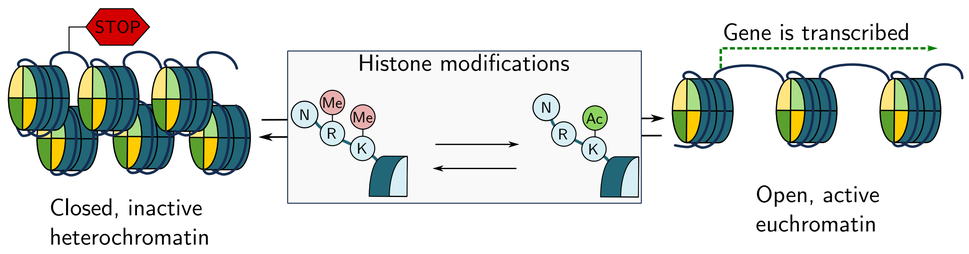
Introduction

Biological Background



- Post-translational modifications of the amino-terminal histone tails play an important role in regulating DNA accessibility.
- Histone modifcations are specfic for genomic activity, e.g., H3K27ac (**ac**etylation of lysine (**K**) at position **27** in the amino acid sequence of the histone protein **H3**) is enriched in active promoters and enhancers and trimethylated H3K9 is enriched in heterochromatic regions.

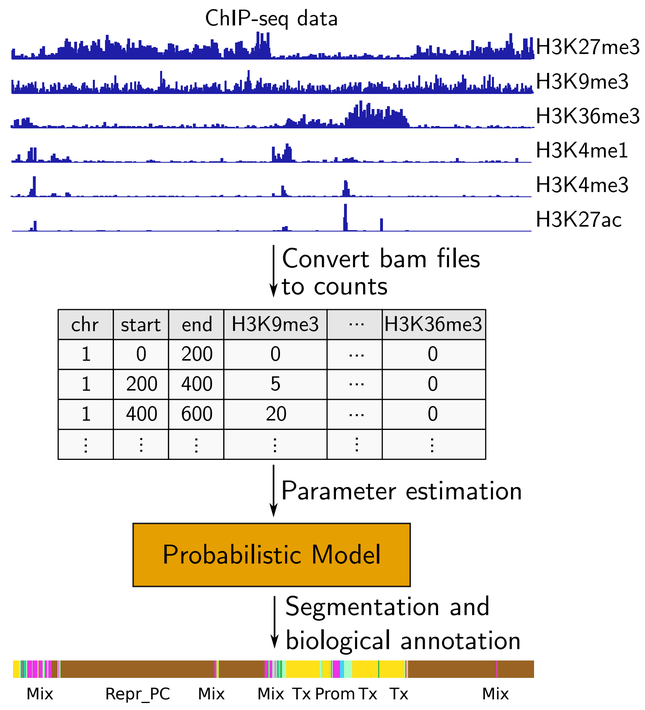
Thus, the combinatorial enrichment or depletion of histone marks defines chromatin states with distinct functional roles.

Chromatin Segmentation

Segmentation tools enable the automated discovery of chromatin states using a probabilistic model that captures histone patterns and adjacency relations via emission and transition probabilities. Since the segmentation quality depends on the modeling assumptions, we developed in EpiSegMix a more flexible hidden Markov model (HMM) [1].

Existing Tools

- ChromHMM: HMM with independent Bernoulli emissions [2]
- EpiCSeg: HMM with negative multinomial emissions [3]
- Segway: Bayesian network with Gaussian mixture emissions [4]



Method

Emission Modeling

ChIP-seq data has varying degrees of overdispersion and skewness, requiring flexible distributions to model the emission probabilities. In EpiSegMix, the emission probabilities are modeled as the product of **independent discrete distributions**, allowing a different distribution type to be selected for each histone mark.
Supported distributions: **Poisson, Binomial, Beta Binomial, Negative Binomial, Beta Negative Binomial, Sichel**

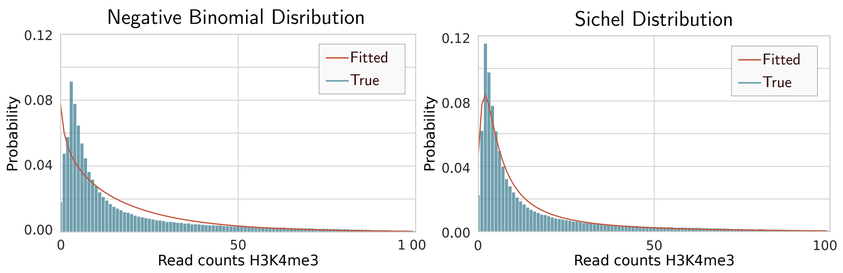
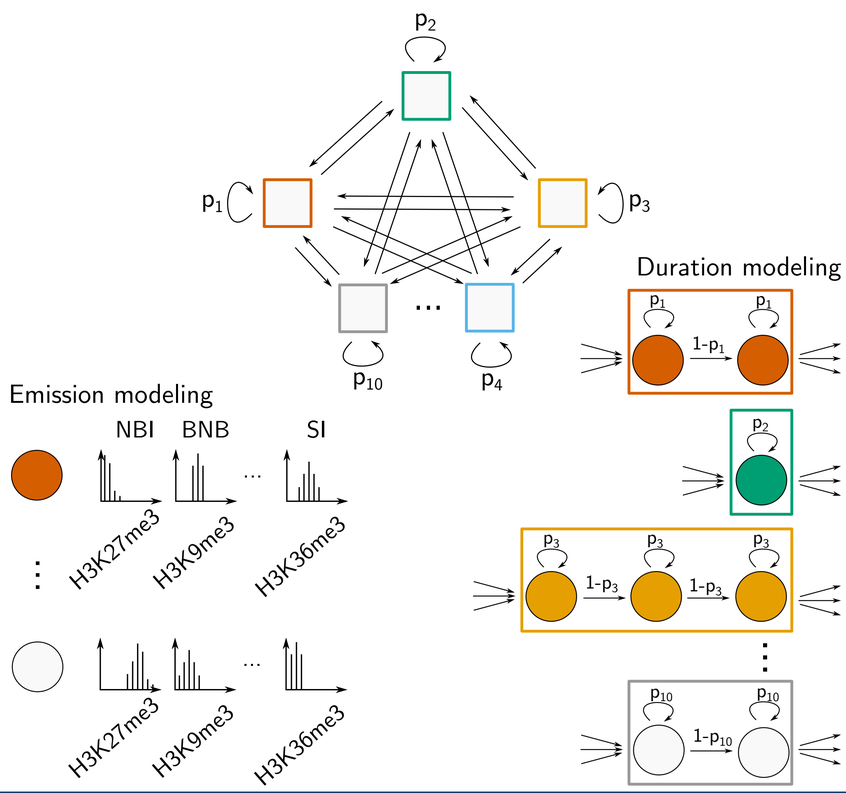


Figure: Emission distribution for H3K4me3 (in H3K4me3 enriched state, HepG2).

Probabilistic Model of EpiSegMix



Duration Modeling

Histone marks are enriched as either narrow peaks or broad domains depending on their biological function, which necessitates flexible modeling of various state durations. In EpiSegMix, each state in the HMM is internally represented as a sub-HMM with a **left-right topology**, resulting in a state duration that follows a **Negative Binomial** distribution instead of a Geometric distribution.

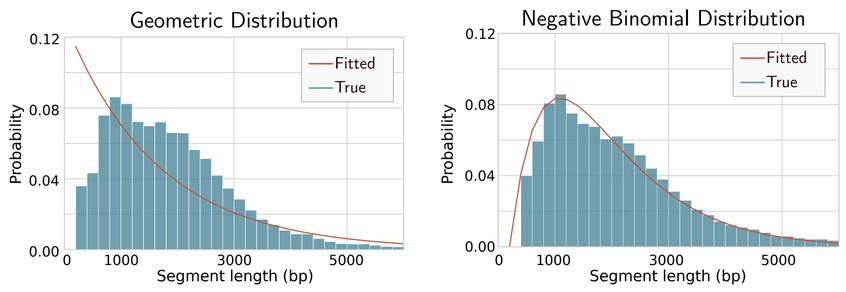
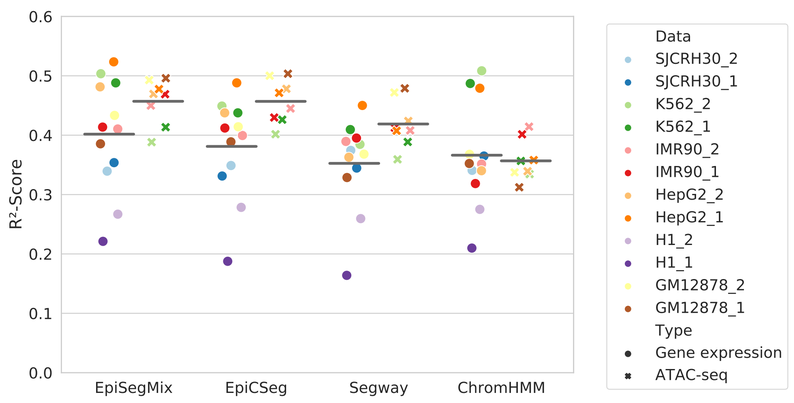


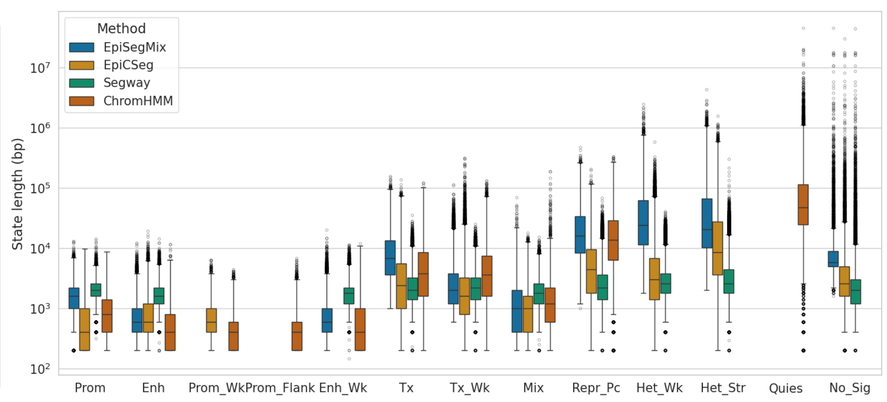
Figure: State duration distribution for a classic (Geometric) and extended-state (Negative Binomial) topology (for HepG2 promoter state).

Results

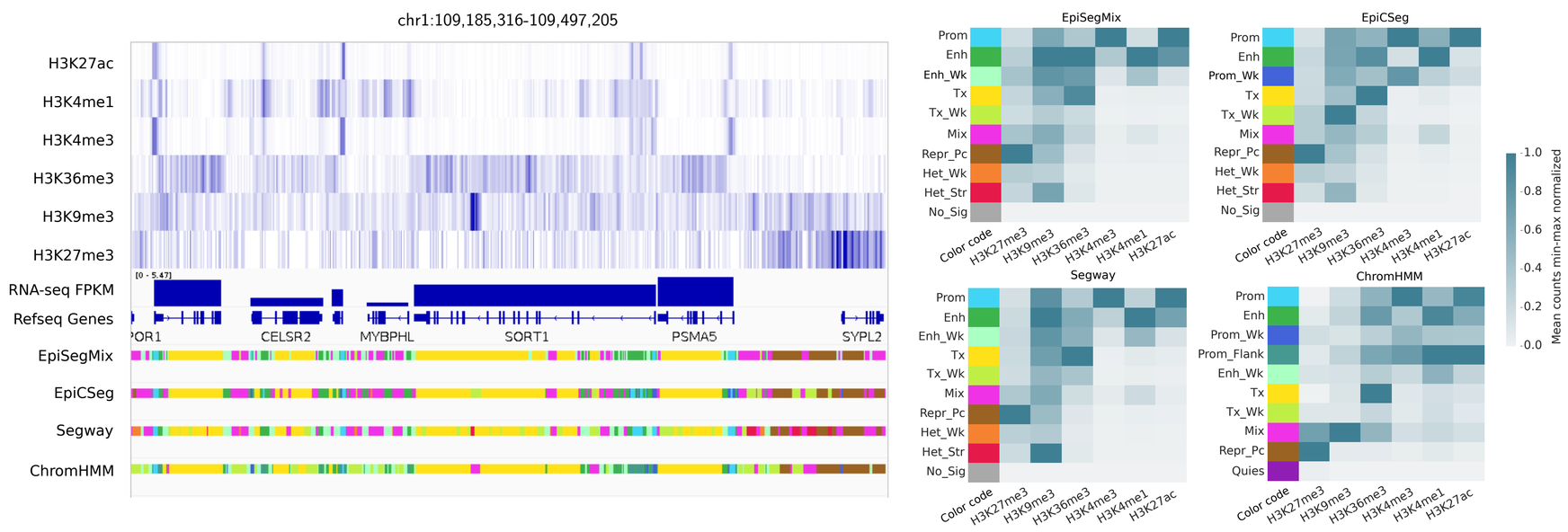
- EpiSegMix is most predictive of gene expression and ATAC-seq counts.
- Flexible duration modeling allows to accurately capture both short and long segments.
- All methods find similar histone mark patterns of activating and repressing marks.



(a) Prediction of gene expression and ATAC-Seq counts using state labels as categorical predictors.



(b) State length distribution in base pairs across biologically labelled states for the different methods (for K562_1).



(c) IGV screenshot with histone signals, RNA-seq and segmentation tracks. The heatmaps show the histone enrichment per state (for K562_1).

Figure: Comparison between EpiSegMix and EpiCSeg, Segway and ChromHMM (on publicly available ENCODE data).

References

[1] Johanna Elena Schmitz, Nihit Aggarwal, Lukas Laufer, Jörn Walter, Abdulrahman Salhab, and Sven Rahmann. EpiSegMix: a flexible distribution hidden Markov model with duration modeling for chromatin state discovery. *Bioinformatics*, 40(4), April 2024.

[2] Jason Ernst and Manolis Kellis. Discovery and characterization of chromatin states for systematic annotation of the human genome. *Nature Biotechnology*, 28(8):817–825, August 2010.

[3] Alessandro Mammana and Ho-Ryun Chung. Chromatin segmentation based on a probabilistic model for read counts explains a large portion of the epigenome. *Genome Biology*, 16(1):151, July 2015.

[4] Rachel C W Chan, Maxwell W Libbrecht, Eric G Roberts, Jeffrey A Bilmes, William Stafford Noble, and Michael M Hoffman. Segway 2.0: Gaussian mixture models and minibatch training. *Bioinformatics*, 34(4):669–671, February 2018.

Availability

EpiSegMix GitLab

Publication

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